Session 5 – Lab

Contact Filter Console Application

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Course: Android App Development using Gen AI
Module: Kotlin Fundamentals

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1. Objective

The objective of this lab is to build a small console-based Contact Management program using Kotlin. This exercise focuses on practicing Kotlin fundamentals such as data classes, nullability, conditionals, lambda expressions, higher-order functions, and formatted printing.

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2. Problem Statement

Develop a console application that:

- Stores a list of contacts.
- Allows filtering contacts using lambda expressions.
- Handles nullable phone and email fields.
- Prints formatted contact information.
- Demonstrates filtering for:
    - All contacts
    - Favorite contacts
    - Contacts with email

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3. Kotlin Concepts Used

- Data Classes
- Nullable Types (`String?`)
- Elvis Operator (`?:`)
- Higher-Order Functions
- Lambda Expressions
- List Filtering (`filter()`)
- String Templates

- Conditional Expressions

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4. Source Code

```
// ----------------------------------
// Contact Filter Console Application
// ----------------------------------

data class Contact(
val name: String,     val
phone: String?,     val
email: String?,     val
favorite: Boolean
)

fun formatContact(c: Contact): String {
val phoneDisplay = c.phone ?: "N/A"
   val emailDisplay = c.email ?: "N/A"

   return "${c.name} | Phone: $phoneDisplay | Email: $emailDisplay${favOrEmpty(c.favorite)}"
}

fun favOrEmpty(isFavorite: Boolean): String {
return if (isFavorite) " | Favorite" else ""
}

fun filterContacts(     contacts:
List<Contact>,     predicate:
(Contact) -> Boolean
): List<Contact> {
   return contacts.filter(predicate)
}

fun printContacts(title: String, contacts: List<Contact>) {
println(title)

   contacts.forEachIndexed { index, contact ->
println("${index + 1}. ${formatContact(contact)}")
   }

   println()
}

fun main() {

   val allContacts = listOf(
      Contact("Maya Patel", "+91-90000-11111", "maya@example.com", true),
      Contact("Raj Sharma", null, "raj@sample.com", false),
      Contact("Li Wei", "+86-13800000000", null, false),
      Contact("Sara Ali", null, null, false),
      Contact("Tom OBrien", "+1-555-1234", "tom@example.org", true)
   )
```

```
    printContacts("--- All Contacts ---", allContacts)

    val favoriteContacts = filterContacts(allContacts) { it.favorite }     printContacts("-
-- Favorite Contacts ---", favoriteContacts)

    val contactsWithEmail = filterContacts(allContacts) { it.email != null }
printContacts("--- Contacts with Email ---", contactsWithEmail)
}
```

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5. Sample Output

```
--- All Contacts ---
1. Maya Patel | Phone: +91-90000-11111 | Email: maya@example.com | Favorite
2. Raj Sharma | Phone: N/A | Email: raj@sample.com
3. Li Wei | Phone: +86-13800000000 | Email: N/A
4. Sara Ali | Phone: N/A | Email: N/A
5. Tom OBrien | Phone: +1-555-1234 | Email: tom@example.org | Favorite

--- Favorite Contacts ---
1. Maya Patel | Phone: +91-90000-11111 | Email: maya@example.com | Favorite
2. Tom OBrien | Phone: +1-555-1234 | Email: tom@example.org | Favorite

--- Contacts with Email ---
1. Maya Patel | Phone: +91-90000-11111 | Email: maya@example.com | Favorite
2. Raj Sharma | Phone: N/A | Email: raj@sample.com
3. Tom OBrien | Phone: +1-555-1234 | Email: tom@example.org | Favorite
```

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6. Explanation

The program defines a Contact data class with nullable phone and email fields. The formatContact function handles null values using the Elvis operator and formats contact information properly.

The filterContacts function demonstrates higher-order functions and lambda usage by filtering the contact list based on conditions such as favorite status or availability of email.

The printContacts function prints formatted contact data with numbering.

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7. Conclusion

This lab strengthened understanding of Kotlin fundamentals such as null safety, lambdas, filtering collections, and formatted output. It improved problem-solving ability using functional programming concepts in Kotlin.

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